

IS6400 Business Data Analytics

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Lecture 3

☐ Data Types

☐ Descriptive Analytics

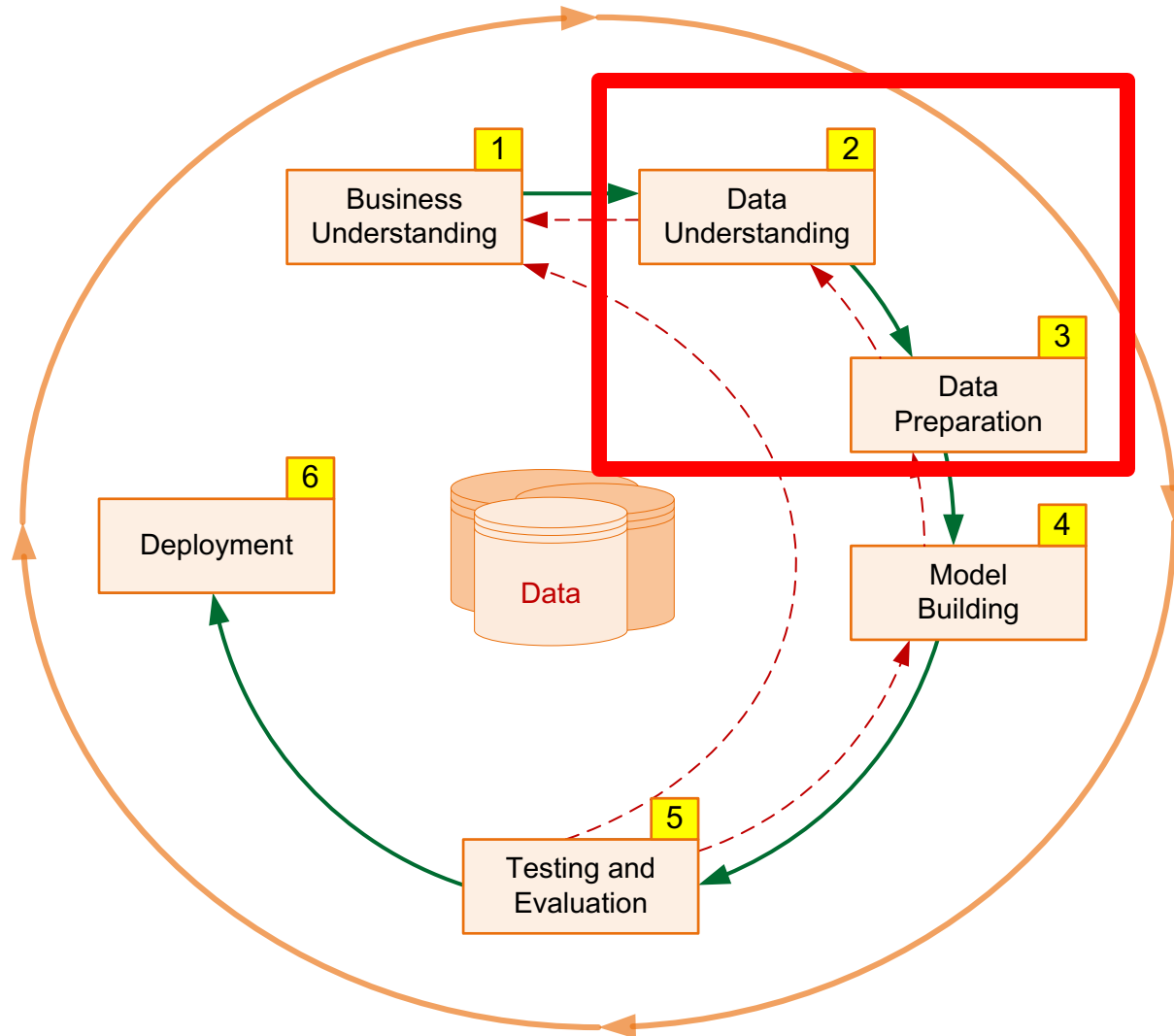
- ☐ Simple Statistics

- ☐ Data Processing

Reading Material:

- ☐ DM Chapter 3

Data



Data

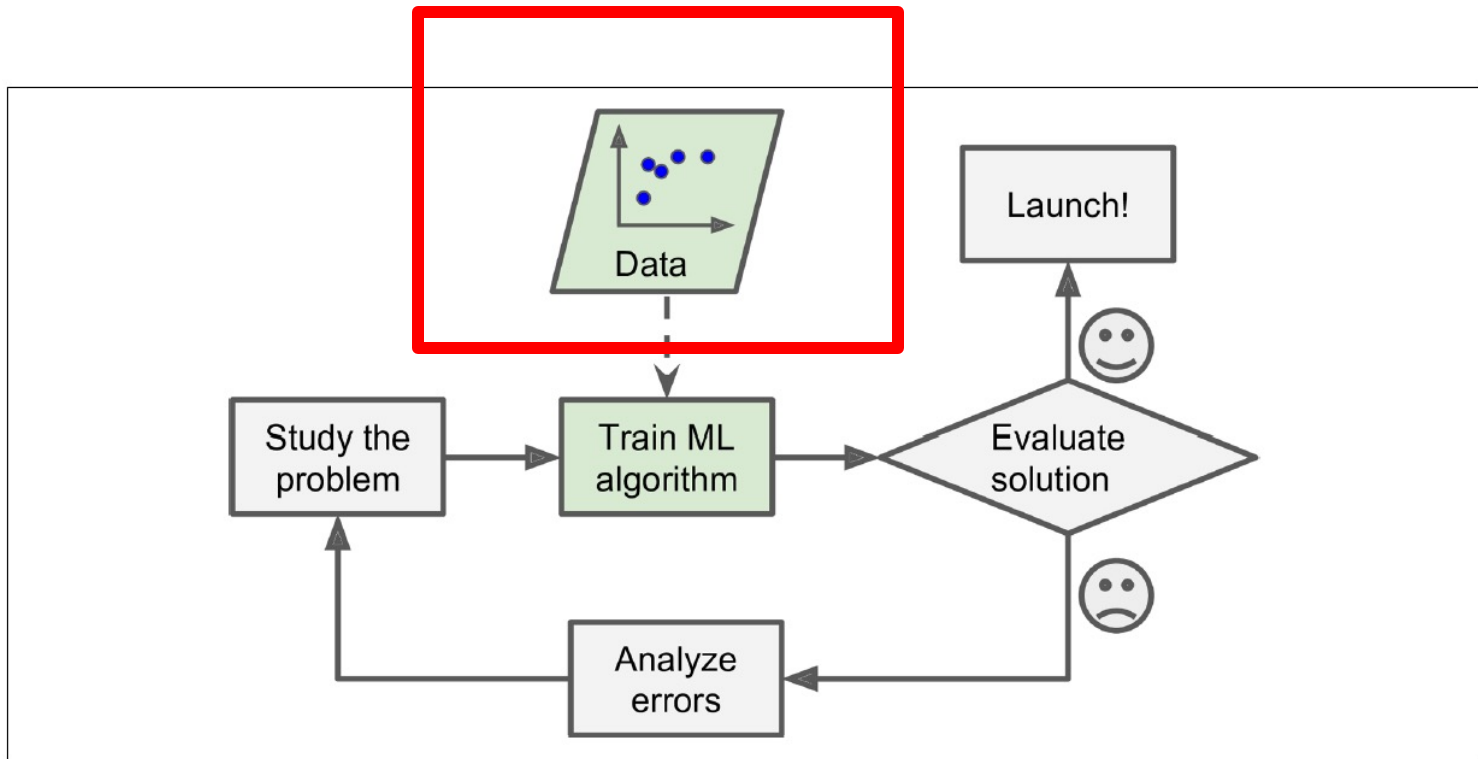


Figure 1-2. Machine Learning approach

What is Data?

- ❖ Collection of data **objects** and their **attributes**
- ❖ An attribute is a property or characteristic of an object
 - ∞ Examples: eye color of a person, temperature, etc.
 - ∞ Attribute is also known as variable, field, characteristic, or feature
- ❖ A collection of attributes describe an object
 - ∞ Object is also known as record, point, case, sample, entity, or instance

Attributes

Objects

Tid	Refund	Marital Status	Taxable Income	Cheat
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

Data – Airbnb – Price Analysis

```
import pandas as pd
ResearchData=pd.read_csv('Airbnb.csv') #load data as pandas dataframe
ResearchData.head()
```

	id	log_price	property_type	accommodates	bathrooms	bed_type	cancellation_policy
0	6901257	5.010635	Apartment	3	1.0	Real Bed	strict
1	6304928	5.129899	Apartment	7	1.0	Real Bed	strict
2	7919400	4.976734	Apartment	5	1.0	Real Bed	moderate
3	13418779	6.620073	House	4	1.0	Real Bed	flexible
4	3808709	4.744932	Apartment	2	1.0	Real Bed	moderate

Data – Supermarket Sales Analysis

- You work for a supermarket, which has an ERP that documents the companies' daily transactions. Your boss wants to know the operation status of the company. You are assigned to give him a report.



Customer information

Store information

Transaction information

Purchase Date	Customer ID	Gender	Marital Status	Homeowner	Children	Annual Income	City	State or Province	Country	Product Family	Product Department	Product Category	Units Sold	Revenue
12/18/2007	7223	F	S	Y	2	\$30K - \$50K	Los Angeles	CA	USA	Food	Snack Foods	Snack Foods	5	\$27.38
12/20/2007	7841	M	M	Y	5	\$70K - \$90K	Los Angeles	CA	USA	Food	Produce	Vegetables	5	\$14.90
12/21/2007	8374	F	M	N	2	\$50K - \$70K	Bremerton	WA	USA	Food	Snack Foods	Snack Foods	3	\$5.52
12/21/2007	9619	M	M	Y	3	\$30K - \$50K	Portland	OR	USA	Food	Snacks	Candy	4	\$4.44
12/22/2007	1900	F	S	Y	3	\$130K - \$150K	Beverly Hills	CA	USA	Drink	Beverages	Carbonated Beverages	4	\$14.00
12/22/2007	6696	F	M	Y	3	\$10K - \$30K	Beverly Hills	CA	USA	Food	Deli	Side Dishes	3	\$4.37
12/23/2007	9673	M	S	Y	2	\$30K - \$50K	Salem	OR	USA	Food	Frozen Foods	Breakfast Foods	4	\$13.78
12/25/2007	354	F	M	Y	2	\$150K +	Yakima	WA	USA	Food	Canned Foods	Canned Soup	6	\$7.34
12/25/2007	1293	M	M	Y	3	\$10K - \$30K	Bellingham	WA	USA	Non-Consumable	Household	Cleaning Supplies	1	\$2.41
12/25/2007	7938	M	S	N	1	\$50K - \$70K	San Diego	CA	USA	Non-Consumable	Health and Hygiene	Pain Relievers	2	\$8.96
12/26/2007	9357	F	M	N	0	\$30K - \$50K	Beverly Hills	CA	USA	Food	Snack Foods	Snack Foods	3	\$11.82
12/26/2007	3097	M	M	Y	1	\$30K - \$50K	Beverly Hills	CA	USA	Food	Baking Goods	Baking Goods	5	\$14.45
12/26/2007	2741	M	S	N	3	\$70K - \$90K	Bellingham	WA	USA	Food	Canned Foods	Canned Tuna	4	\$19.18
12/26/2007	2032	F	M	N	3	\$10K - \$30K	Yakima	WA	USA	Non-Consumable	Household	Plastic Products	4	\$19.50
12/27/2007	6651	M	S	N	0	\$30K - \$50K	Portland	OR	USA	Food	Produce	Fruit	5	\$13.06
12/27/2007	5220	M	M	Y	2	\$30K - \$50K	Salem	OR	USA	Non-Consumable	Health and Hygiene	Household	5	\$12.43

Attribute Values

- ❖ Attribute values are numbers or symbols assigned to an attribute for a particular object
- ❖ Distinction between attributes and attribute values
 - ⌘ Same attribute can be mapped to different attribute values
 - ❖ Example: height can be measured in feet or meters
 - ⌘ Different attributes can be mapped to the same set of values
 - ❖ Example: Attribute values for ID and age are integers
 - ❖ But properties of attribute values can be different

Types of Attributes

❖ There are different types of attributes

❧ Nominal

- ❖ Examples: ID numbers, eye color, zip codes
- ❖ Nominal values provide only enough information to distinguish one object from another.

❧ Ordinal

- ❖ Examples: rankings (e.g., taste of potato chips on a scale from 1-10), grades, height in {tall, medium, short}
- ❖ The values of an ordinal attribute provide enough information to order objects.

Types of Attributes

- ❖ There are different types of attributes

❧ Interval

- ❖ Examples: calendar dates, temperatures in Celsius or Fahrenheit.
- ❖ For interval attributes, **the differences between values are meaningful. (no ratios)**

❧ Ratio

- ❖ Examples: temperature in Kelvin, length, time, counts
- ❖ For ratio variables, **both differences and ratios are meaningful.**

Properties of Attribute Values

❖ The type of an attribute depends on which of the following properties it possesses:

∞ Distinctness: $= \neq$

∞ Order: $< >$

∞ Addition: $+ -$

∞ Multiplication: $* /$

∞ Nominal attribute: distinctness

∞ Ordinal attribute: distinctness & order

∞ Interval attribute: distinctness, order & addition

∞ Ratio attribute: all 4 properties

Discrete and Continuous Attributes

❖ Discrete Attribute

- ⌘ Has only a finite or countably infinite set of values
- ⌘ Examples: zip codes, counts, or the set of words in a collection of documents
- ⌘ Often **represented as integer variables**.
- ⌘ Note: **binary attributes** are a special case of discrete attributes

❖ Continuous Attribute

- ⌘ Has **real numbers as attribute values**
- ⌘ Examples: temperature, height, or weight.
- ⌘ Practically, real values can only be measured and represented using a finite number of digits.
- ⌘ Continuous attributes are typically represented as floating-point variables.

Asymmetric Nominal Attributes

- ❖ Only presence (a non-zero attribute value) is regarded as important
- ❖ Examples:
 - ⌘ Student Attendance? (attend: 1, absent: 0)
 - ⌘ Student 1 and student 2 come together?
 - ❖ (student 1 attendance, student 2 attendance)
 - ❖ (1,1)->come together, (1,0)->only student 1 comes, (0,1)->only student 2 comes, (0,0) both are absent
 - ⌘ N student come together? We need a vector of N binaries.
- ❖ We need two asymmetric binary attributes to represent one ordinary binary attribute
 - ⌘ Association analysis uses asymmetric attributes

Example

<i>Tid</i>	Refund	Marital Status	Taxable Income	Cheat
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

Tid: Nominal

Refund: Nominal

Marital: Nominal

Income: Ratio

Types of data sets

❖ Record

- ❧ Data Matrix

- ❧ Document Data

- ❧ Transaction Data

❖ Graph

- ❧ World Wide Web

- ❧ Molecular Structures

❖ Ordered

- ❧ Spatial Data

- ❧ Temporal Data

- ❧ Sequential Data

- ❧ Genetic Sequence Data

Data Matrix

- ❖ If data objects have the same fixed set of numeric attributes, then the data objects can be thought of as points in a multi-dimensional space, where each dimension represents a distinct attribute
- ❖ Such data set can be represented by an m by n matrix, where **there are m rows, one for each object, and n columns, one for each attribute**

Projection of x Load	Projection of y load	Distance	Load	Thickness
10.23	5.27	15.22	2.7	1.2
12.65	6.25	16.22	2.2	1.1

Document Data

- ❖ Each document becomes a 'term' vector,
 - ⌘ each term is a component (attribute) of the vector,
 - ⌘ the value of each component is the number of times the corresponding term occurs in the document.

	team	coach	play	ball	score	game	win	lost	timeout	season
Document 1	3	0	5	0	2	6	0	2	0	2
Document 2	0	7	0	2	1	0	0	3	0	0
Document 3	0	1	0	0	1	2	2	0	3	0

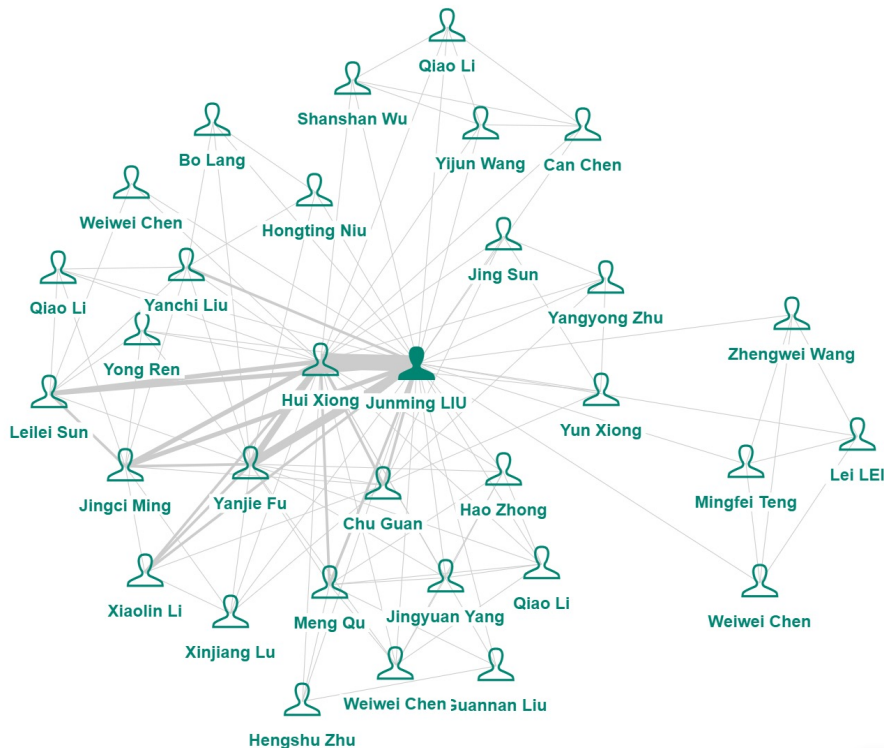
Transaction Data

- ❖ A special type of record data, where
 - ↻ each record (transaction) involves a set of items.
 - ↻ For example, consider a grocery store. The set of products purchased by a customer during one shopping trip constitute a transaction, while the individual products that were purchased are the items.

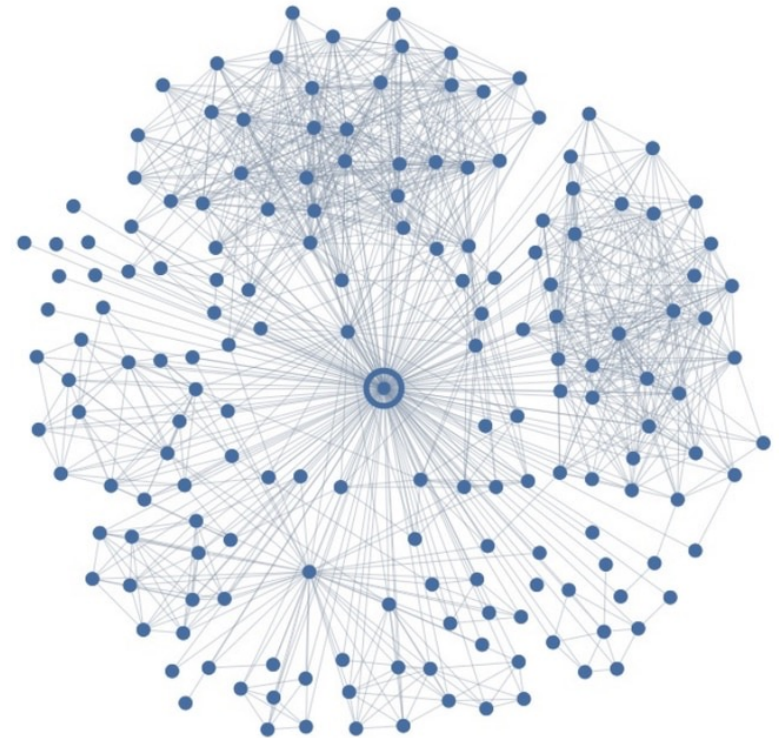
<i>TID</i>	<i>Items</i>
1	Bread, Coke, Milk
2	Beer, Bread
3	Beer, Coke, Diaper, Milk
4	Beer, Bread, Diaper, Milk
5	Coke, Diaper, Milk

Graph Data

❖ Examples: Generic graph, a Molecule, and Webpages



Co-author Map



Facebook Friendship

Data Quality ...

- ❖ What kinds of data quality problems?
- ❖ How can we detect problems with the data?
- ❖ What can we do about these problems?
- ❖ Examples of data quality problems:
 - ❧ Noise and outliers
 - ❧ Missing values
 - ❧ Duplicate data

Important Characteristics of Structured Data

❧ Data Quality

❧ Dimensionality

- ❖ The number of attributes
- ❖ Curse of Dimensionality

❧ Sparsity

- ❖ Only presence counts
- ❖ Less than 1% of the entries are non-zero.

❧ Resolution

- ❖ Patterns depend on the scale
- ❖ The surface of the Earth seems very uneven at a resolution of a few meters, but is relatively smooth at a resolution of tens of kilometers.

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- ☒ **Descriptive Analytics**
 - ☒ **Simple Statistics**
 - ☐ Data Processing

Reading Material:

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Descriptive Analytics

- ❖ The first target of descriptive analytics is to present data in a form that makes sense to people so that we can further contemplate about the data and get information to support decision making.
 - ❧ A tendency to race through “descriptive statistics”
 - ❧ The “descriptive statistics” is **simple, complex and important.**
- ❖ Three ways to explore a data set:
 - ❧ Numerical summary measures,
 - ❖ Descriptive statistics: for example counts, percentages, averages, and measures of variability.
 - ❖ Tables of summary measures: for example pivot table.
 - ❧ A variety of graphs, including bar charts, pie charts, histograms, scatterplots, and time series graphs (data visualization)

Measures for Non-Numerical Variables

- ❖ Only a few possibilities exist for describing categorical variables, based on **counting**.
 - ↻ Count based on the number of categories
 - ↻ Count the number of observations in each category
 - ↻ Frequency distribution: frequency of observations in non-overlapping classes/cells
 - ↻ Excel's COUNTIF function.
 - ❖ The function takes two arguments: The data range (where) and a criterion that one wants to count (what).

Statistics for Numerical Variables

❖ Single Variable

❧ Measures of **central tendency**

- ❖ Mean, Median, Mode
- ❖ Minimum, Maximum, Percentiles, and Quartiles

❧ Measures of **dispersion/variability**

- ❖ Range
- ❖ IQR: interquartile range
- ❖ Variance
- ❖ Standard deviation

❧ Measures of distribution **shape**

- ❖ Skewness: occurs when the sample is lack of symmetry
- ❖ Kurtosis: this is all about the extreme observations.

❖ Multiple variables

❧ **Covariance and correlation.**

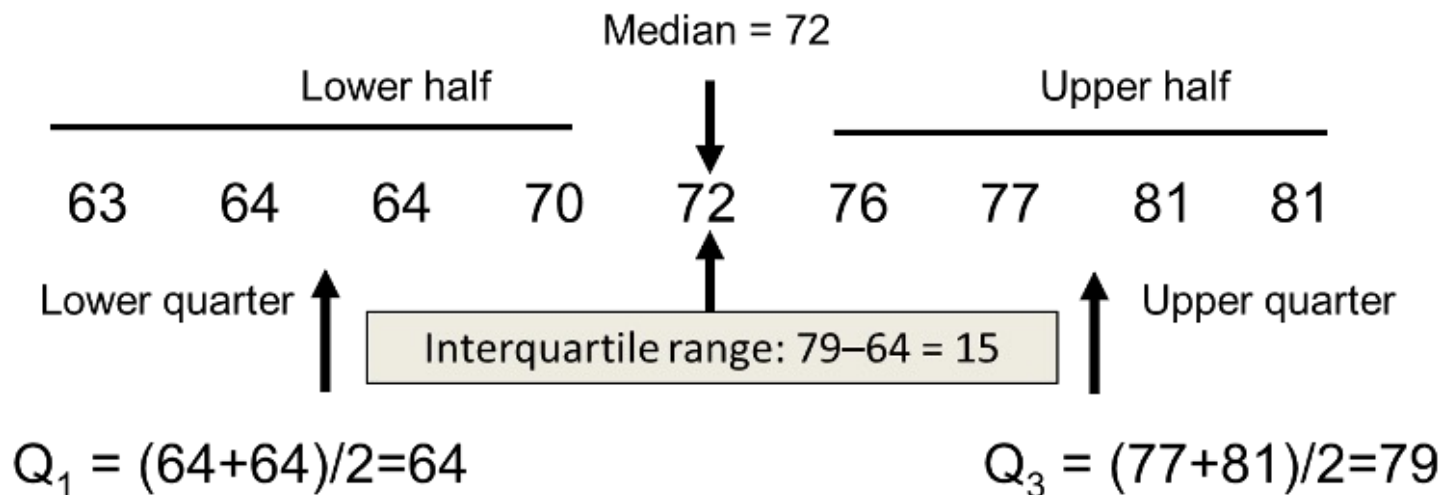
Percentiles and Quartiles

For any percentage p , the p th **percentile** is the value such that a percentage p of all values are less than it.

- ❖ The **quartiles** divide the data into four groups, each with (approximately) a quarter of all observations.
 - ∞ The first, second and third quartiles are the percentiles corresponding to $p = 25\%$, $p = 50\%$, and $p = 75\%$.
 - ∞ The second quartile ($p = 50\%$) is equal to the median.

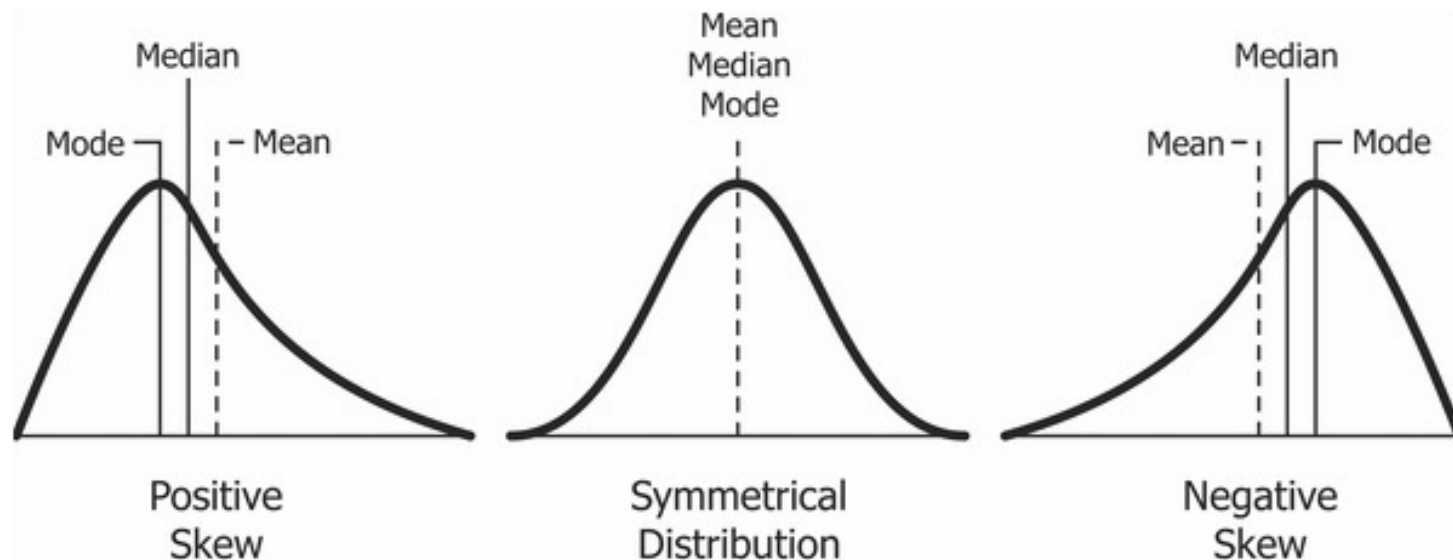
Measures of Dispersion: Range

- ❖ The **range** is the maximum value minus the minimum value.
- ❖ The **interquartile range (IQR)** is the third quartile minus the first quartile.
 - ∞ Thus, it is the range of the middle 50% of the data.
 - ∞ It is less sensitive to extreme values than the range.



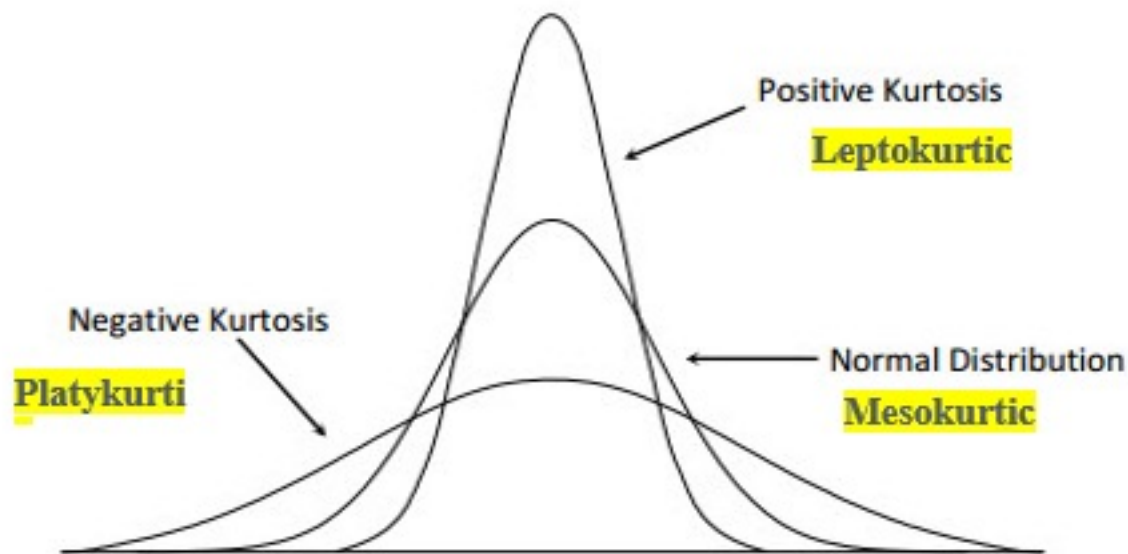
Measures of Distribution Shape

- ❖ Skewness: indicates relative symmetry



Kurtosis

❖ Kurtosis: indicates tailedness



Measures of Association

- ❖ The relationship between two or more variables.
- ❖ Example:
Data from 49 top liberal arts and research universities can be used to answer questions:
 - ❧ Is Top 10% HS **related** to Graduation %?
 - ❧ Is Acceptance Rate **related** to Expenditures/Student?
 - ❧ Is Median SAT **related** to Acceptance Rate?

	A	B	C	D	E	F	G
1	Colleges and Universities						
2							
3	School	Type	Median SAT	Acceptance Rate	Expenditures/Student	Top 10% HS	Graduation %
4	Amherst	Lib Arts	1315	22%	\$ 26,636	85	93
5	Barnard	Lib Arts	1220	53%	\$ 17,653	69	80
6	Bates	Lib Arts	1240	36%	\$ 17,554	58	88
7	Berkeley	University	1176	37%	\$ 23,665	95	68
8	Bowdoin	Lib Arts	1300	24%	\$ 25,703	78	90
9	Brown	University	1281	24%	\$ 24,201	80	90
10	Bryn Mawr	Lib Arts	1255	56%	\$ 18,847	70	84

Covariance

- ❖ A measure of the linear association between two (continuous) variables, X and Y .
- ❖ For a population, COVARIANCE.P:

$$\text{cov}(X, Y) = \frac{\sum_{i=1}^N (x_i - \mu_x)(y_i - \mu_y)}{N}$$

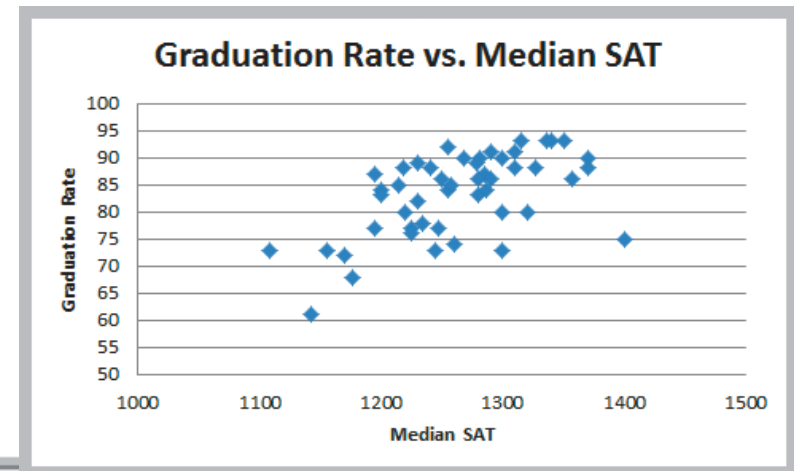
- ❖ For a sample, COVARIANCE.S:

$$\text{cov}(X, Y) = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{n - 1}$$

Covariance Example

❖ Scatterplot of the *Colleges and Universities* data

$$\text{cov}(X, Y) = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{n - 1}$$



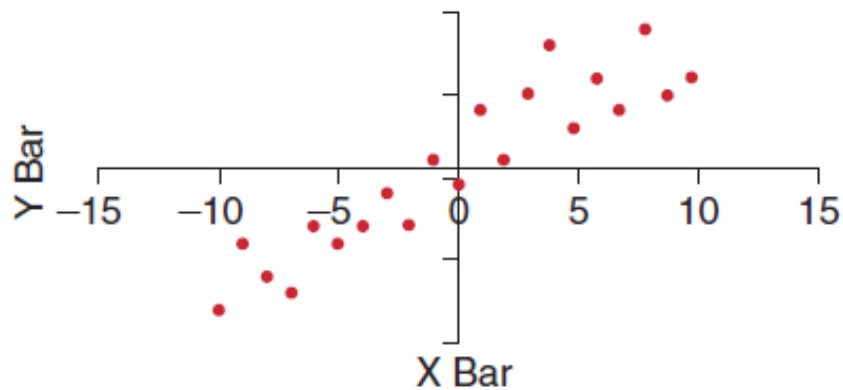
	A	B	C	D	E	F
1		Graduation % (X)	Median SAT (Y)	X - Mean(X)	Y - Mean(Y)	(X - Mean(X))(Y - Mean(Y))
2		93	1315	9.755	51.898	506.2698875
3		80	1220	-3.245	-43.102	139.8617243
4		88	1240	4.755	-23.102	-109.8525614
46		78	1234	-5.245	-29.102	152.6372345
47		86	1250	2.755	-13.102	-36.09745939
48		91	1290	7.755	26.898	208.5964182
49		93	1336	9.755	72.898	711.1270304
50		93	1350	9.755	86.898	847.698459
51	Mean	83.245	1263.102		Sum	12641.77551
52					Count	49
53					Covariance	263.3703231
54						
55					COVARIANCE.S	263.3703231

Correlation

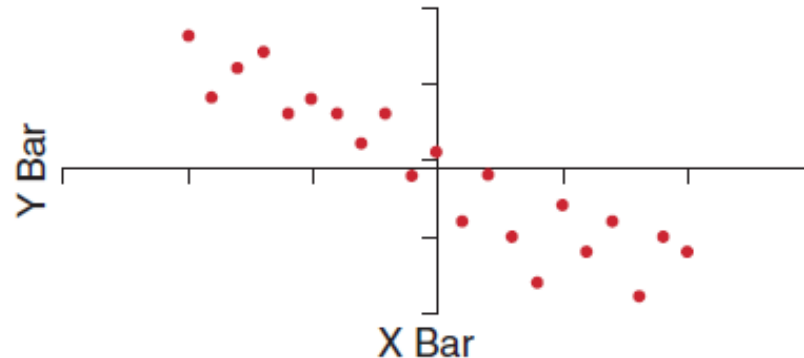
- ❖ A measure of the linear association between two variables, X and Y .
- ❖ Unlike covariance, the scale of variation of each variable does not influence correlation.
- ❖ The correlation value is always between -1 and +1.

$$\rho_{xy} = \frac{\text{cov}(X, Y)}{\sigma_x \sigma_y}$$

Correlation

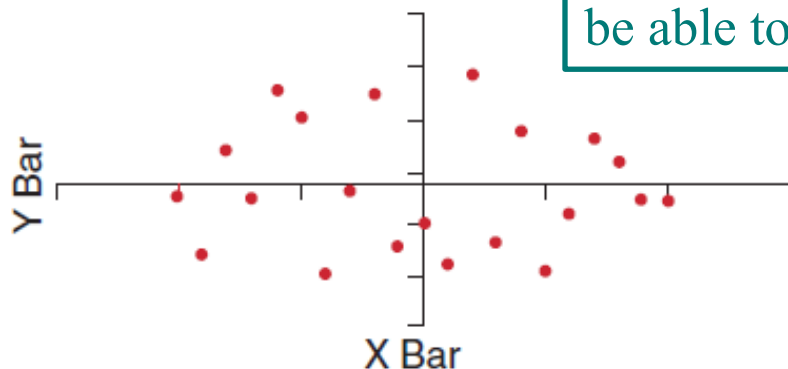


(a) Positive Correlation

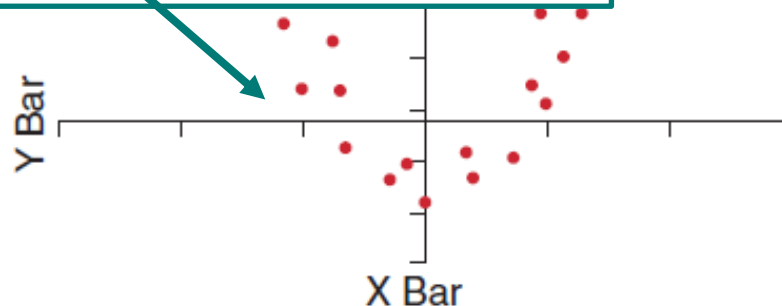


(b) Negative Correlation

Correlation is linear concept, it may not be able to recover nonlinear relationships.



(c) No Correlation



(d) A Nonlinear Relationship with No Linear Correlation

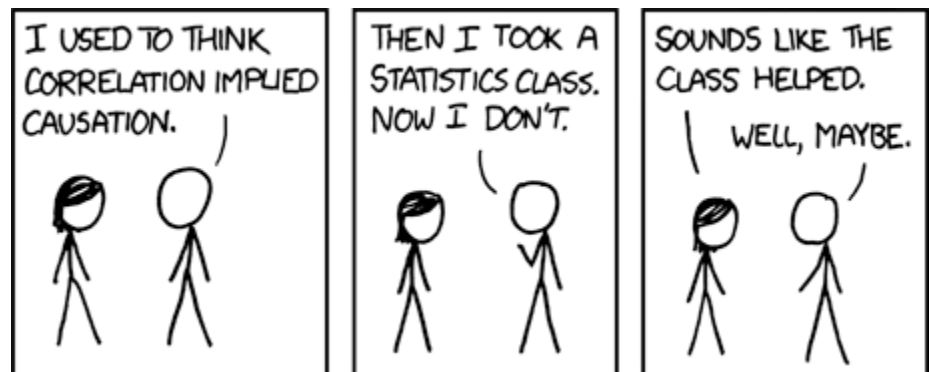
Correlation $\neq$ Causality

Careful!

- ❖ We may see positive correlation between the number of bank branches in a region and the total deposit amount, does that mean opening more branches will lead to more deposit?
- ❖ Causality is more important but is challenging to establish
- ❖ Beyond analytics

- 🌀 Experiment

- 🌀 Quasi-experiment



Association Rule Mining

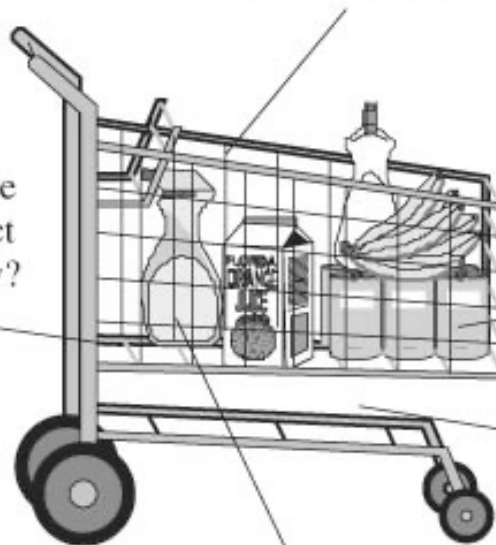
❖ Customers would buy items together.

☞ Ex. Milk + bread.

In this shopping basket, the shopper purchased a quart of orange juice, some bananas, dish detergent, some window cleaner, and a six pack of soda.

How do the demographics of the neighborhood affect what customers buy?

Is soda typically purchased with bananas? Does the brand of soda make a difference?



What should be in the basket but is not?

Are window cleaning products purchased when detergent and orange juice are bought together?

Association Rule Mining

- ❖ Definition: Given a set of transactions, find rules that will predict the occurrence of an item based on the occurrences of other items in the transaction

Market-Basket transactions

<i>TID</i>	<i>Items</i>
1	Bread, Milk
2	Bread, Diaper, Beer, Eggs
3	Milk, Diaper, Beer, Coke
4	Bread, Milk, Diaper, Beer
5	Bread, Milk, Diaper, Coke

Example of Association Rules

$\{\text{Diaper}\} \rightarrow \{\text{Beer}\},$
 $\{\text{Milk, Bread}\} \rightarrow \{\text{Eggs, Coke}\},$
 $\{\text{Beer, Bread}\} \rightarrow \{\text{Milk}\},$

Implication means co-occurrence,
not causality!

Various Associations

❖ Between values

☞ e.g. Apple $\Rightarrow$ Coke

❖ Between categories of values

☞ e.g. Food $\Rightarrow$ Magazine

❖ Between values of attributes

☞ e.g. Married:yes $\Rightarrow$ OwnHouse:yes

❖ Over time period

☞ e.g. year 1: Database $\Rightarrow$ year 2: Data Mining

Why Do We Care?

❖ Product placement

- Whole foods: next to flowers are birthday cards
- Wal-Mart customers who purchase Barbie dolls have a 60% likelihood of also purchasing one of three types of candy bars

❖ Recommendations

- As you are looking at iPad, you might want cases or a keyboard attachment

❖ Bundling

- E.g. travel “packages” – flight, hotel, car

❖ Other applications

- Website / catalog design
- Medical complications

Basic Concepts

❖ Itemset

↪ A collection of one or more items

❖ Example: {Milk, Bread, Diaper}

↪ k-itemset

❖ An itemset that contains k items

❖ Support count (σ)

↪ Frequency of occurrence of an itemset

↪ E.g. $\sigma(\{\text{Milk, Bread, Diaper}\}) = 2$

❖ Support

↪ Fraction of transactions that contain an itemset

↪ E.g. $s(\{\text{Milk, Bread, Diaper}\}) = 2/5$

❖ Frequent Itemset

↪ An itemset whose support is greater than or equal to a *minsup* threshold

A transaction (TID) may contain a list of purchased items (basket of items)

<i>TID</i>	<i>Items</i>
1	Bread, Milk
2	Bread, Diaper, Beer, Eggs
3	Milk, Diaper, Beer, Coke
4	Bread, Milk, Diaper, Beer
5	Bread, Milk, Diaper, Coke

Definition: Association Rule

- Association Rule

- An implication expression of the form $X \rightarrow Y$, where X and Y are itemsets
- Example:
 $\{\text{Milk, Diaper}\} \rightarrow \{\text{Beer}\}$

<i>TID</i>	<i>Items</i>
1	Bread, Milk
2	Bread, Diaper, Beer, Eggs
3	Milk, Diaper, Beer, Coke
4	Bread, Milk, Diaper, Beer
5	Bread, Milk, Diaper, Coke

- Rule Evaluation Metrics

- Support (s)
 - ◆ Fraction of transactions that contain both X and Y
- Confidence (c)
 - ◆ Measures how often items in Y appear in transactions that contain X

Example:

$$\{\text{Milk, Diaper}\} \Rightarrow \{\text{Beer}\}$$

$$s = \frac{\sigma(\text{Milk, Diaper, Beer})}{|T|} = \frac{2}{5} = 0.4$$

$$c = \frac{\sigma(\text{Milk, Diaper, Beer})}{\sigma(\text{Milk, Diaper})} = \frac{2}{3} = 0.67$$

Support

$$\text{support} = \frac{\# \text{ of transactions containing all items in antecedent and consequent}}{\# \text{ of transactions in database}}$$

<i>TID</i>	<i>Items</i>
1	Bread, Milk, Diapers
2	Bread, Diapers
3	Bread, Beer
4	Milk, Eggs, Shoes

- ❖ Support for {bread} → {diapers} is 2/4
- ❖ In other words, 50% of transactions include this pair of items

- ❖ Support quantifies the significance of the **co-occurrence** of the items involved in a rule
- ❖ In practice, we only care about item sets with strong enough support

Measuring The Performance Of Association Rules

- ❖ Two popular measures: **confidence** and **lift**

$$\text{confidence} = \frac{\text{\# of transactions containing items in antecedent and consequent}}{\text{\# of transactions containing items in antecedent}}$$

<i>TID</i>	<i>Items</i>
1	Bread, Milk, Diapers
2	Bread, Diapers
3	Bread, Beer
4	Milk, Eggs, Shoes

- ❖ Confidence for {bread} → {diapers} is 2/3
- ❖ In other words, conditional on that the basket contains bread, there is probability 2/3 that the same basket also contains diapers

Valid Association Rules

- ❖ A rule has to meet a minimum support and a minimum confidence
 - Both thresholds determined by the decision maker
- ❖ Why need both threshold?
 - Example of a rule with high confidence & low support:
 - A cell phone company database contains all call destinations for each account
 - {Germany} $\rightarrow$ {France, Belgium} with confidence = 100%
 - Support is 1 of 100K accounts

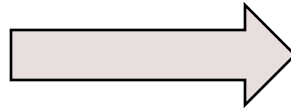
Valid Association Rules

❖ Suppose we use:

- Minimum support: 50%
- Minimum confidence: 50%

❖ Check support:

<i>TID</i>	<i>Items</i>
1	Bread, Milk, Diapers
2	Bread, Diapers
3	Bread, Beer
4	Milk, Eggs, Shoes



<i>Frequent Pattern</i>	<i>Support</i>
{Bread}	75%
{Milk}	50%
{Diapers}	50%
{Bread, Diapers}	50%

❖ Check rules – only two survives:

- Bread → Diapers (Support = 50%, Confidence = 66.67%)
- Diapers → Bread (support = 50%, confidence = 100%)

Implications Of Support & Confidence

- ❖ A rule with **low support** is typically not interesting from business point of view as it may **simply occur by chance**
- ❖ **High confidence** represents **strong reliability** of the inference made by a rule
 - The higher the value, it is more likely than people buy X will more likely buy Y than not buy Y
 - However, association rules do not imply causality but how strong is the co-occurrence among the considered items

Caveat About Confidence

❖ Consider: Tea → Coffee

	<i>Coffee</i>	<i>NOT Coffee</i>	
<i>Tea</i>	15	5	20
<i>NOT Tea</i>	75	5	80
	90	10	100

❖ Confidence = $\#(\text{Coffee \& Tea}) / \#(\text{Tea}) = 15/20 = 75\%$

➤ i.e. the probability that someone who has bought tea will also buy coffee, is 75%

❖ Seems good?

Caveat About Confidence

❖ Consider: Tea → Coffee

	<i>Coffee</i>	<i>NOT Coffee</i>	
<i>Tea</i>	15	5	20
<i>NOT Tea</i>	75	5	80
	90	10	100

- ❖ Recall that Confidence = $\#(\text{Coffee} \& \text{Tea}) / \#(\text{Tea}) = 15/20 = 75\%$
- ❖ But, Support or $P(\text{Coffee}) = \#(\text{Coffee}) / 100 = 90 / 100 = 90\%$
 - i.e. the probability that someone would have bought coffee is 90%
 - So, given that tea has been bought, the probability of buying coffee has dropped in the same transaction
 - Although confidence is high, rule is misleading!
- ❖ In fact, the confidence of “NOT Tea → Coffee” is $75/80 = 93.8\%$
- ❖ We are looking for a rule that Confidence($X \rightarrow Y$) > Support(Y)
 - i.e. Confidence($X \rightarrow \text{Coffee}$) > Support(Coffee)

Another Performance Measure: Lift

- ❖ The lift of a rule measures how much more likely the consequent is, given the antecedent

$$\text{lift} = \frac{\text{confidence of the rule}}{\text{support of the consequent}} = \frac{P(Y | X)}{P(Y)}$$

	<i>Coffee</i>	<i>NOT Coffee</i>	
<i>Tea</i>	15	5	20
<i>NOT Tea</i>	75	5	80
	90	10	100

Tea → Coffee

- ❖ Confidence is 75%
- ❖ Support of Coffee is 90%
- ❖ Lift = $0.75 / 0.9 = 0.8333 < 1 \rightarrow$ this rule is worse than not having any rule (negatively associated)
 - < 1 : substitution effect > 1 : dependence effect
 - How much our confidence has increased that coffee will be purchased given that tea was purchased

Exercise On Support: Phone Faceplates

<i>TID</i>	<i>Items</i>
1	Bread, Milk
2	Bread, Diaper, Beer, Eggs
3	Milk, Diaper, Beer, Coke
4	Bread, Milk, Diaper, Beer
5	Bread, Milk, Diaper, Coke

Example:

$\{\text{Milk, Diaper}\} \Rightarrow \{\text{Beer}\}$

$$s = \frac{\sigma(\text{Milk, Diaper, Beer})}{|T|} = \frac{2}{5} = 0.4$$

$$c = \frac{\sigma(\text{Milk, Diaper, Beer})}{\sigma(\text{Milk, Diaper})} = \frac{2}{3} = 0.67$$

IS6400 Business Data Analytics

Dr. LIU Junming

City University of Hong Kong

Lecture 3

- ☐ Data Types
- ☐ Descriptive Analytics
 - ☐ Simple Statistics
 - ☒ Data Processing

Reading Material:

- ☐ DM Chapter 3

Aggregation

- ❖ Combining two or more attributes (or objects) into a single attribute (or object)

- ❖ Purpose

 - ∞ Data reduction

 - ❖ Reduce the number of attributes or objects

 - ∞ Change of scale

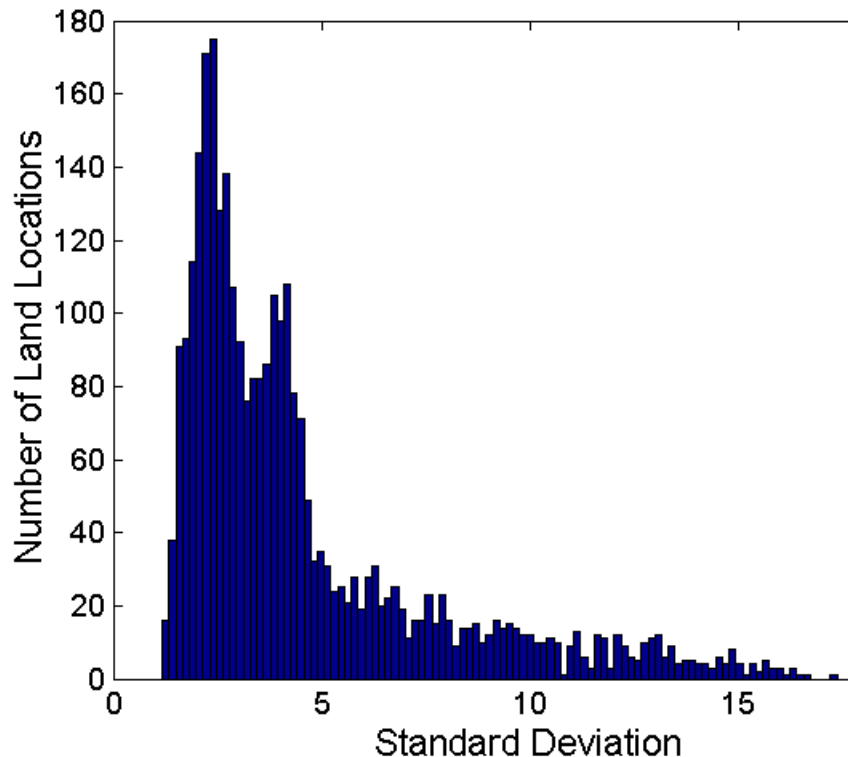
 - ❖ Cities aggregated into regions, states, countries, etc

 - ∞ More “stable” data

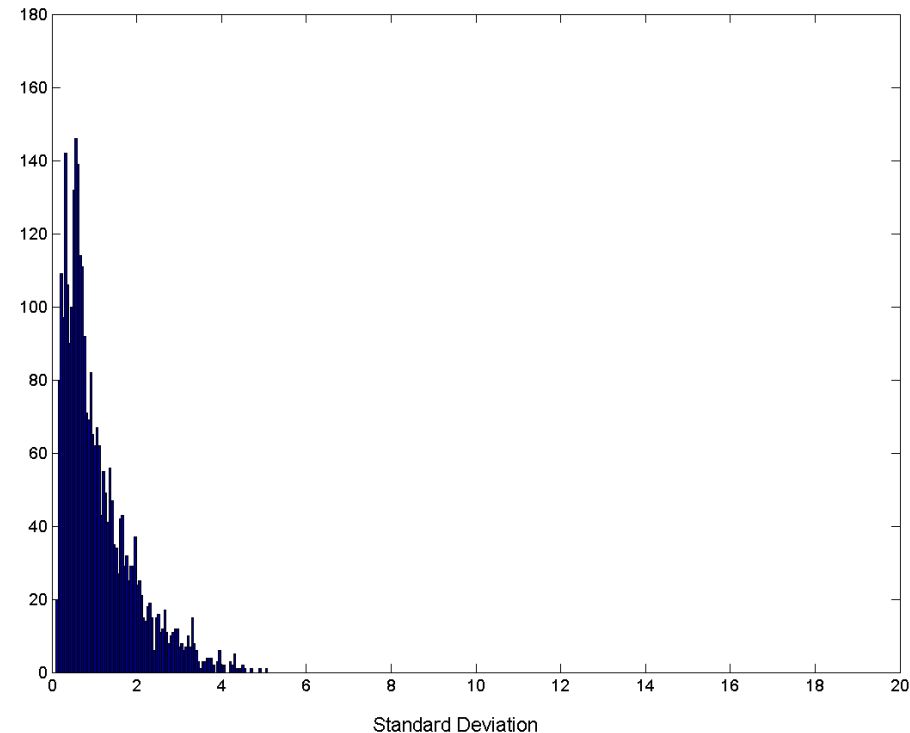
 - ❖ Aggregated data tends to have less variability

Example: Precipitation in Australia ...

Variation of Precipitation in Australia



Standard Deviation of Average Monthly Precipitation



Standard Deviation of Average Yearly Precipitation

Sampling

- ❖ Sampling is the main technique employed for data selection.
 - ∞ It is often used for both the preliminary investigation of the data and the final data analysis.
- ❖ Statisticians sample because **obtaining** the entire set of data of interest is too expensive or time consuming.

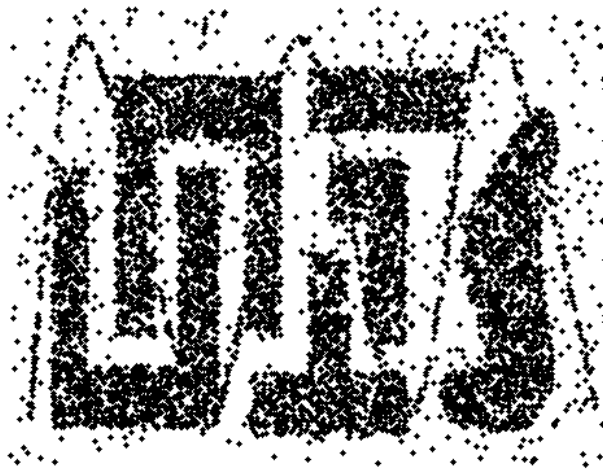
Sampling ...

❖ The key principle for effective sampling is the following:

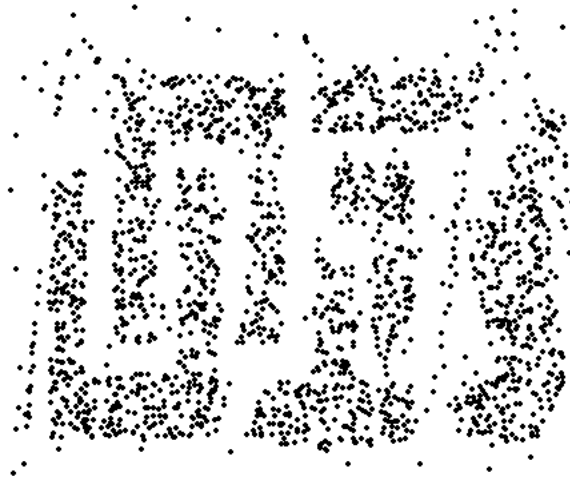
☞ Using a sample will work almost as well as using the entire data sets, if the sample is representative

☞ A sample is representative if it has approximately the same property (of interest) as the original set of data

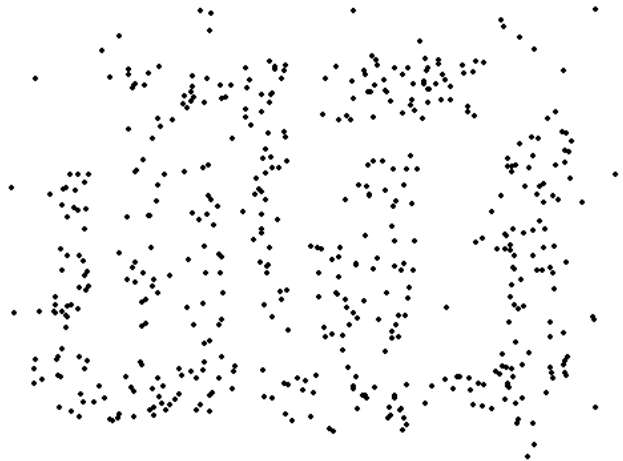
Sample Size



8000 points



2000 Points



500 Points

Types of Sampling

❖ Simple Random Sampling

❧ Sampling without replacement

- ❖ As each item is selected, it is removed from the population

❧ Sampling with replacement

- ❖ Objects are not removed from the population as they are selected for the sample.
 - ❧ In sampling with replacement, the same object can be picked up more than once

❖ Stratified sampling

- ❧ Split the data into several partitions; then draw random samples from each partition

Dimensionality Reduction

❖ Purpose:

- ❧ Avoid curse of dimensionality
- ❧ Reduce amount of time and memory required by data mining algorithms
- ❧ Allow data to be more easily visualized
- ❧ May help to eliminate irrelevant features or reduce noise

❖ Techniques

- ❧ Principal Components Analysis (PCA)
- ❧ Singular Value Decomposition
- ❧ Others: supervised and non-linear techniques

Feature Subset Selection

❖ Redundant features

- ❧ Duplicate much or all of the information contained in one or more other attributes
- ❧ Example: purchase price of a product and the amount of sales tax paid

❖ Irrelevant features

- ❧ Contain no information that is useful for the data mining task at hand
- ❧ Example: students' ID is often irrelevant to the task of predicting students' GPA

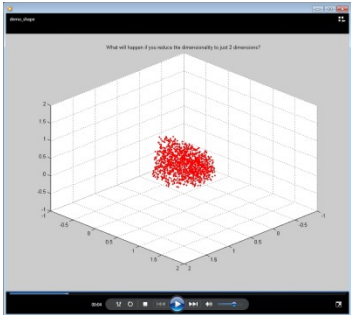
❖ Many techniques developed, especially for classification

Feature Creation

- ❖ Create new attributes that can capture the important information in a data set much more efficiently than the original attributes
- ❖ Three general methodologies:
 - ⌘ Feature extraction
 - ❖ Example: extracting edges from images
 - ⌘ Feature construction
 - ❖ Example: dividing mass by volume to get density
 - ⌘ Mapping data to new space
 - ❖ Example: Fourier and wavelet analysis

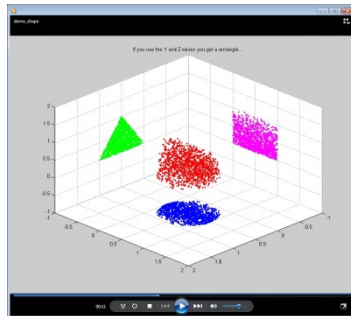
Data Exploration

A cloud of points in 3D

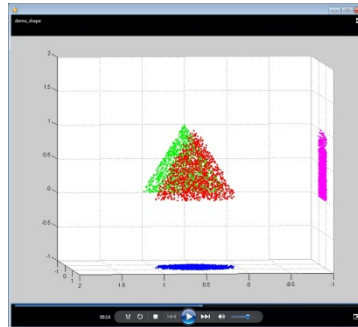


Can be projected into 2D

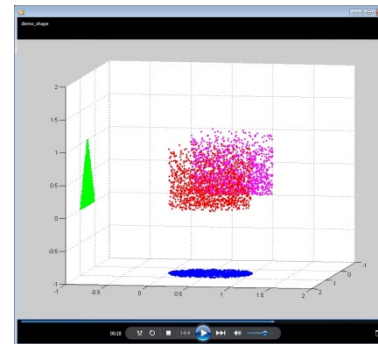
XY or XZ or YZ



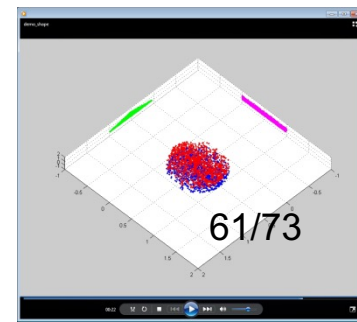
In 2D XZ we see
a triangle



In 2D YZ we see
a square



In 2D XY we see
a circle



Iris Sample Data Set

- ❖ Many of the exploratory data techniques are illustrated with the Iris Plant data set.

- ❧ Can be obtained from the UCI Machine Learning Repository
<http://www.ics.uci.edu/~mlearn/MLRepository.html>

- ❧ From the statistician Douglas Fisher

- ❧ Three flower types (classes)

- ❖ Setosa
- ❖ Virginica
- ❖ Versicolour

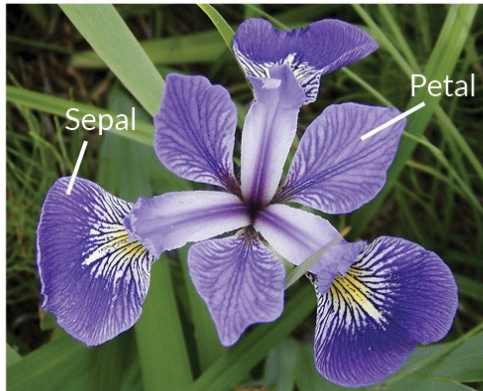
- ❧ Four (non-class) attributes

- ❖ Sepal width and length
- ❖ Petal width and length



Virginica. Robert H. Mohlenbrock. USDA NRCS. 1995. Northeast wetland flora: Field office guide to plant species. Northeast National Technical Center, Chester, PA. Courtesy of USDA NRCS Wetland Science Institute.

Iris Flower



Iris Versicolor



Iris Setosa



Iris Virginica

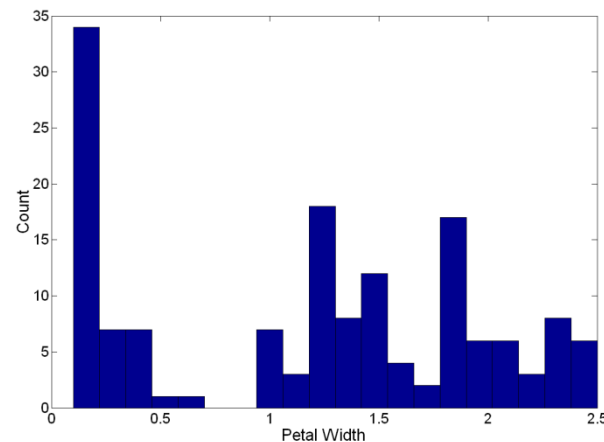
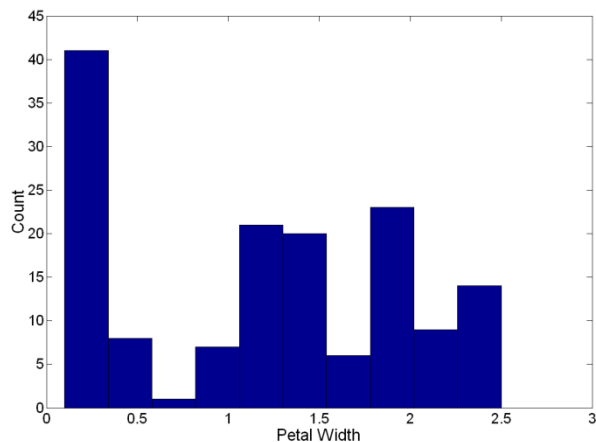
Virginica. Robert H. Mohlenbrock. USDA NRCS. 1995. Northeast wetland flora: Field office guide to plant species. Northeast National Technical Center, Chester, PA. Courtesy of USDA NRCS Wetland Science Institute.

Visualization Techniques: Histograms

❖ Histogram

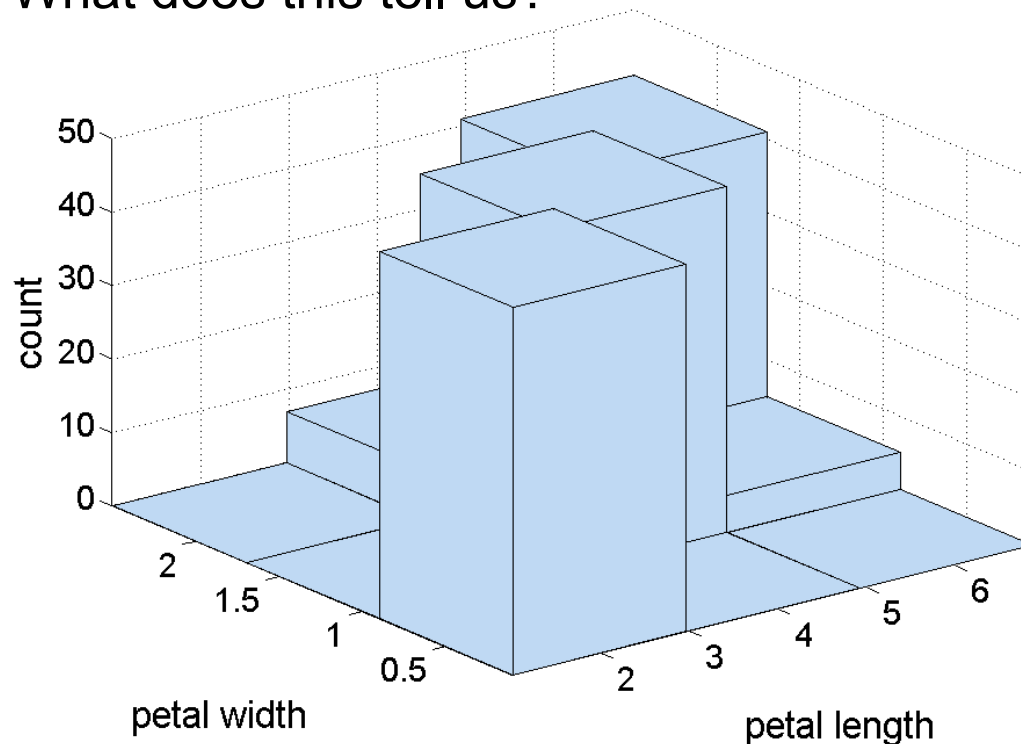
- ☞ Usually shows the distribution of values of a single variable
- ☞ Divide the values into bins and show a bar plot of the number of objects in each bin.
- ☞ The height of each bar indicates the number of objects
- ☞ Shape of histogram depends on the number of bins

❖ Example: Petal Width (10 and 20 bins, respectively)



Two-Dimensional Histograms

- ❖ Show the joint distribution of the values of two attributes
- ❖ Example: petal width and petal length
 - 🔗 What does this tell us?



Visualization Techniques: Box Plots

❖ Box Plots

- ❧ Invented by J. Tukey
- ❧ Another way of displaying the distribution of data
- ❧ Following figure shows the basic part of a box plot

← outlier

← 90th percentile

← 75th percentile

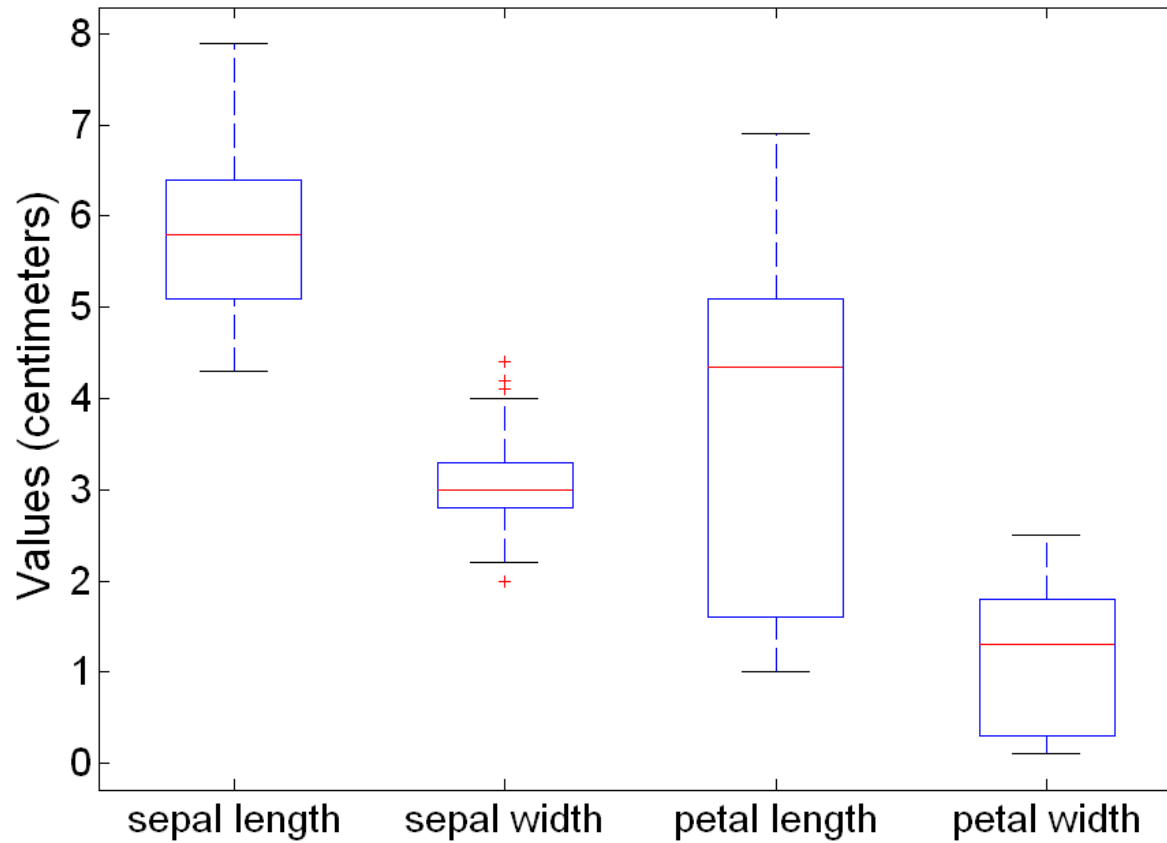
← 50th percentile

← 25th percentile

← 10th percentile

Example of Box Plots

❖ Box plots can be used to compare attributes



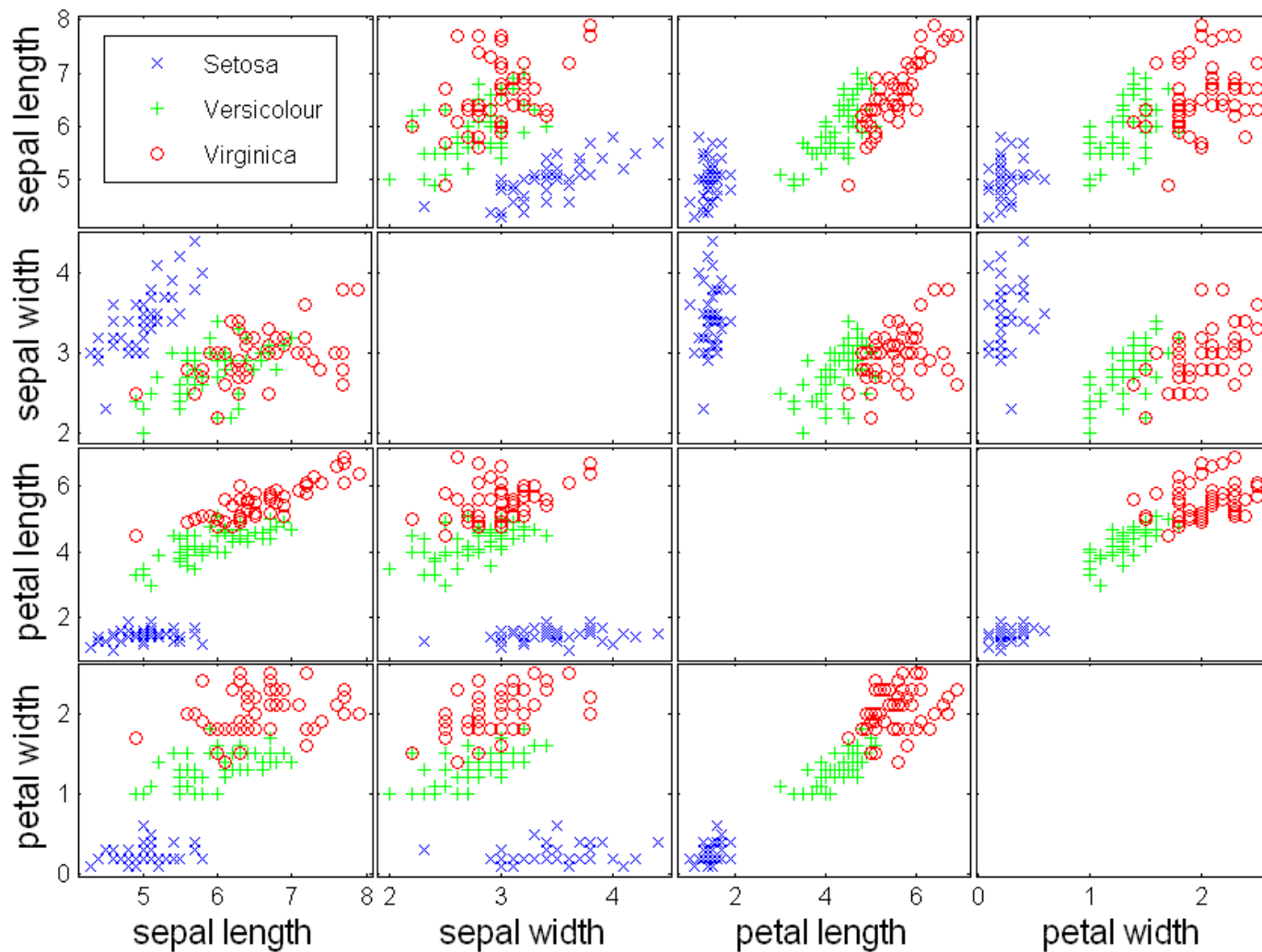
Visualization Techniques: Scatter Plots

❖ Scatter plots

- ❧ Attributes values determine the position
- ❧ Two-dimensional scatter plots most common, but can have three-dimensional scatter plots
- ❧ Often additional attributes can be displayed by using the size, shape, and color of the markers that represent the objects
- ❧ It is useful to have arrays of scatter plots can compactly summarize the relationships of several pairs of attributes

❖ See example on the next slide

Scatter Plot Array of Iris Attributes

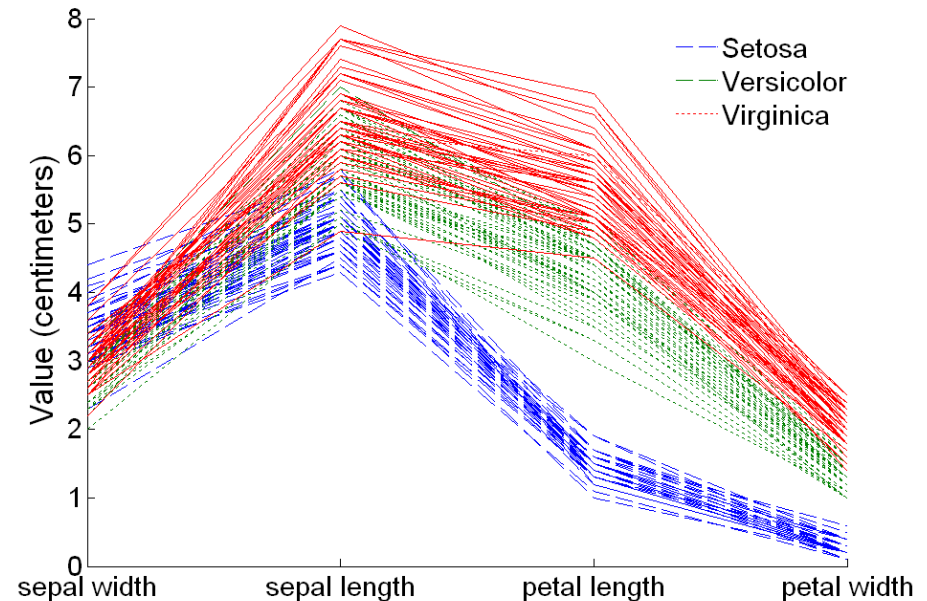
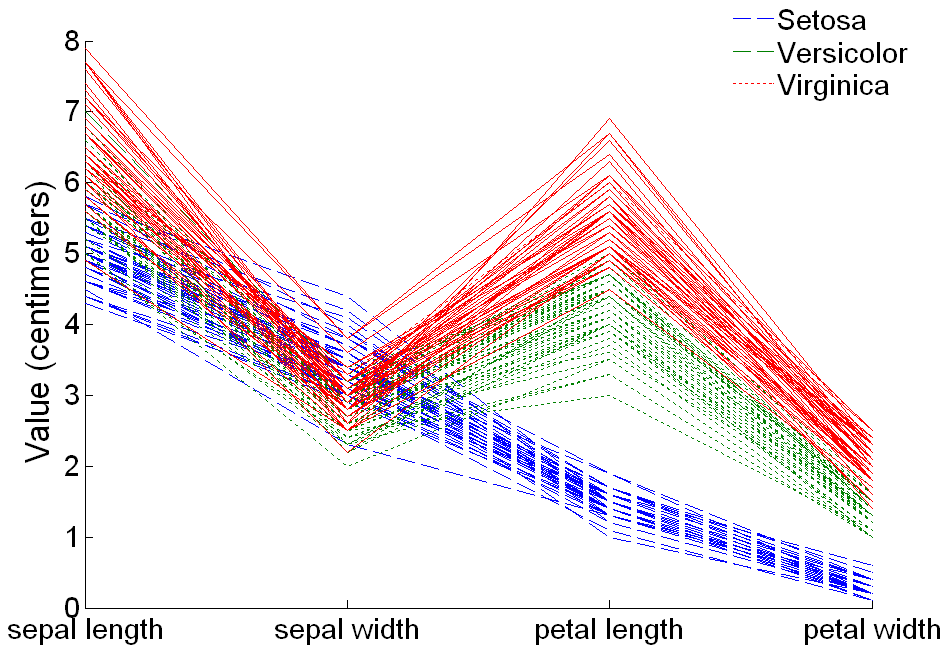


Visualization Techniques: Parallel Coordinates

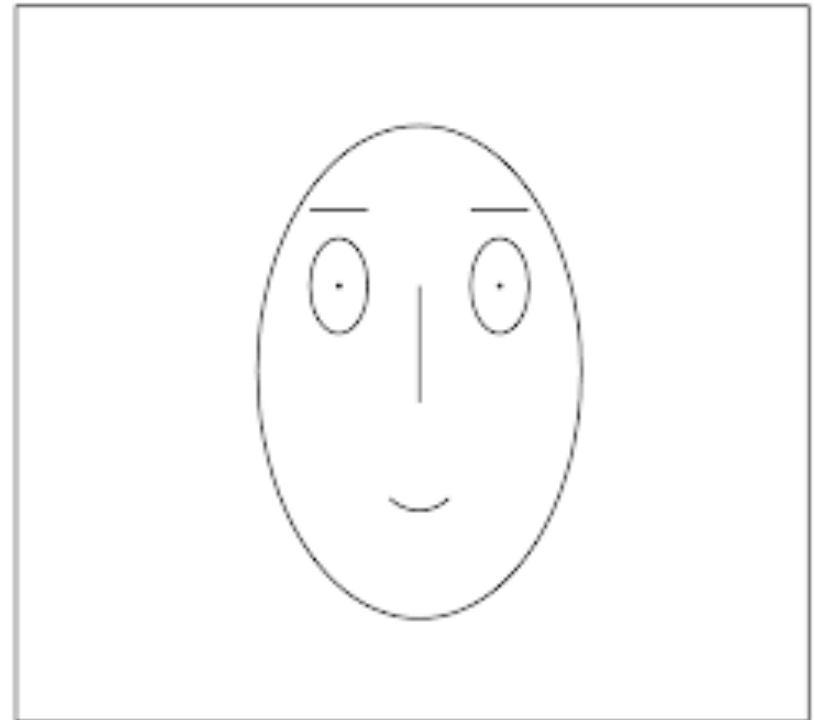
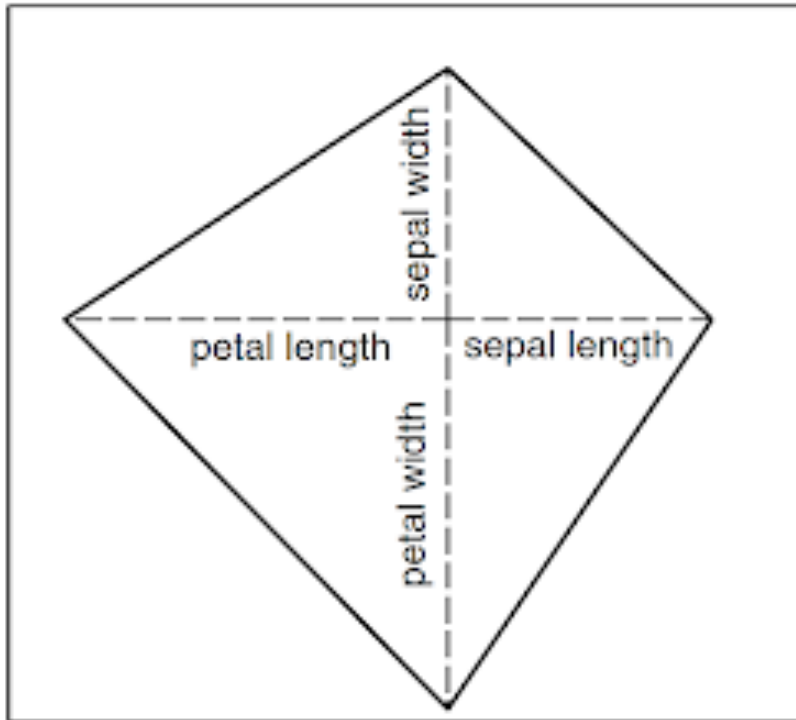
❖ Parallel Coordinates

- ❧ Used to plot the attribute values of high-dimensional data
- ❧ Instead of using perpendicular axes, use a set of parallel axes
- ❧ The attribute values of each object are plotted as a point on each corresponding coordinate axis and the points are connected by a line
- ❧ Thus, each object is represented as a line
- ❧ Often, the lines representing a distinct class of objects group together, at least for some attributes
- ❧ Ordering of attributes is important in seeing such groupings

Parallel Coordinates Plots for Iris Data



Star Plots and Chernoff Faces



Star Plots for Iris Data



1



2



3



4



5

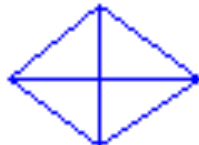
Setosa



51



52



53

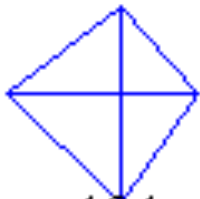


54



55

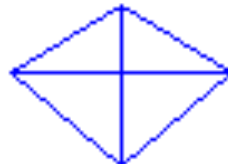
Versicolour



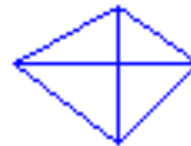
101



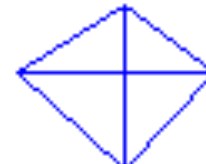
102



103



104



105

Virginica